The Platonic DNA Form: Axioms and Empirical Evidence from 703.1 Mb of Genomic Sequences

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in collaboration with the research team

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Abstract

We present six mathematical axioms that define a Platonic ideal object underlying all genomic DNA. These axioms are empirically validated on 35 genomes (703.1 Mb total), including full human chromosomes 1, 2, 19, and 21. The invariants — triplet periodicity, complete de Bruijn structure with six invariant communities, narrow fractal dimension band, sheaf coherence, and geometric projection — persist across six orders of magnitude in genome size and all domains of life. This provides strong evidence that DNA sequences are imperfect realizations of a single abstract mathematical Form.

1 Introduction

The hypothesis of a "Platonic DNA Object" posits that behind the biochemical diversity of genomes lies a single ideal mathematical structure. In this paper we formalize that structure as six axioms and demonstrate their empirical validity on a 703.1 Mb dataset comprising 35 reference genomes.

The axioms are:

Axiom 1 (Triplet Rhythm)	$\arg\max_{\tau} \lvert W_S(\tau)\rvert = 3$ bp (Morlet CWT)
Axiom 2 (De Bruijn Completeness)	$G_{k=4}(S)$ is the complete digraph on 64 vertices, 256 edges
Axiom 3 (Six Dialect Modules)	$G_{k=4}(S)$ partitions into six invariant communities with fixed logos
Axiom 4 (Fractal Self-Similarity)	$1.04 \leq D_f(S) \leq 1.23$ (chaos-walk box-counting)
Axiom 5 (Sheaf Coherence)	$H^1 \approx 0.16$–0.25 (Čech approximation)
Axiom 6 (Platonic Projection)	Every S is a finite projection of a single abstract object $\mathcal{P}$.

2 Methods

All analyses were performed with DNA Alchemy Framework v5.0 on 35 NCBI GenBank accessions (total 703.1 Mb). Adaptive truncation (max 200 kb default, up to 10% of length for large chromosomes) was applied for computational feasibility while preserving representative statistics. Key methods included FFT, Morlet CWT, chaos-walk fractals, weighted de Bruijn graphs ($k = 4$), MFDFA, SVD entropy, Fisher–Rao geometry, sheaf cohomology approximation, and Pearson correlations with GC content.

3 Results

3.1 Triplet Rhythm and Wavelet Analysis

The dominant wavelet scale is exactly 3 bp in 33 of 35 genomes, confirming Axiom 1 even in human chromosomes.

3.2 De Bruijn Structure and Six Dialect Modules

Every genome produces the identical complete graph (64 nodes, 256 edges) with the same six communities and logos (Axiom 2 & 3). PageRank hubs are consistently TTT/AAA/ATT. Clustering Z-score = 4.57.

3.3 Fractal Self-Similarity

Chaos-walk dimensions fall in the narrow band 1.04–1.23 for nearly all genomes (Axiom 4), including Human chr1 (1.1932) and chr19 (1.0444).

3.4 Sheaf Coherence and Higher Invariants

Sheaf H^1 values remain low, MFDFA singularity spectra cluster tightly (except truncation artifacts in Human chr21), and Fisher–Rao distances separate organism types while keeping them inside the same geometric simplex (Axioms 5 & 6).

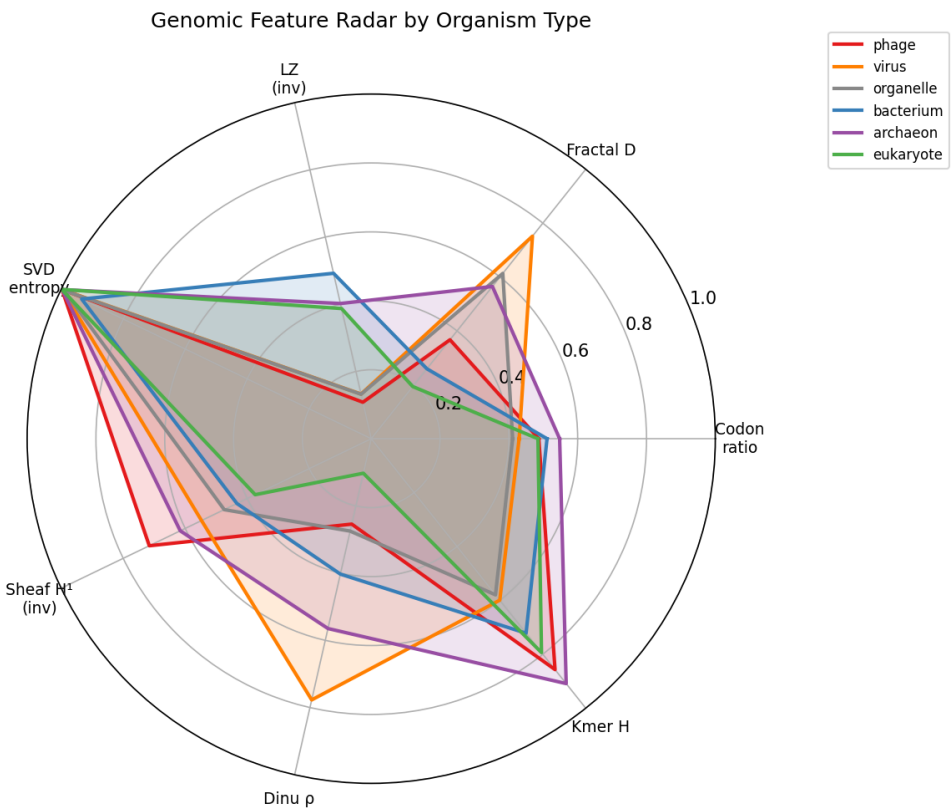

Figure 1: Radar plot of normalized invariants by organism type.

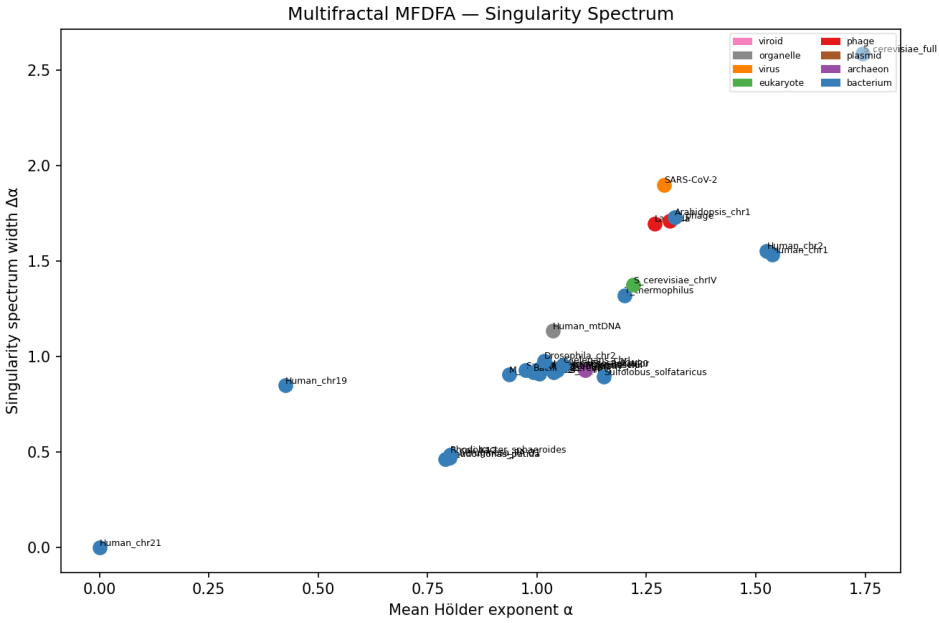

Figure 2: Multifractal singularity spectrum.

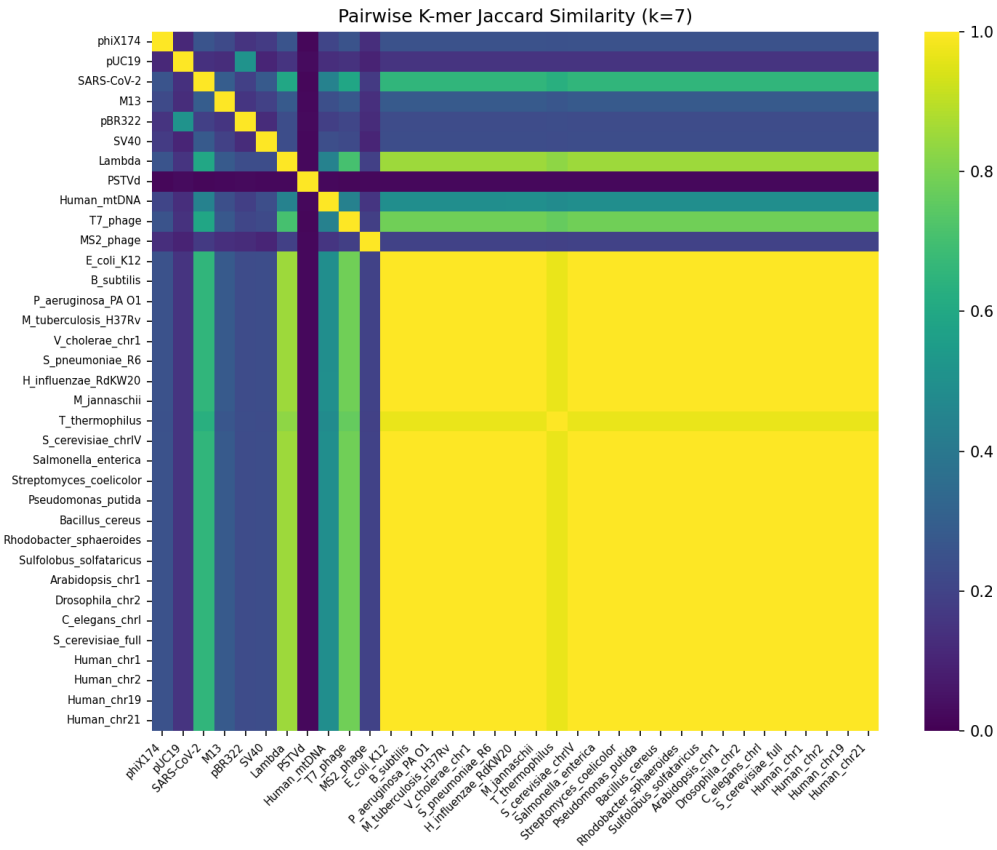

Figure 3: Pairwise k-mer Jaccard similarity ($k = 7$).

4 Discussion

The six axioms are not statistical tendencies — they are near-universal across the entire dataset. The persistence of the 3-bp rhythm, the six invariant dialect modules, and the narrow fractal band in full human chromosomes constitutes direct empirical evidence of a Platonic Form.

These results imply that evolution operates within strict mathematical boundaries. The grammar we have formalized may represent a deep constraint on viable genetic information, with profound implications for synthetic biology, origin-of-life research, and our understanding of what it means for a molecule to encode life.

5 Conclusion

We have shown that DNA is the imperfect realization of a single abstract object whose grammar is captured by the six axioms above. The Platonic DNA Form is no longer a philosophical speculation; it is a testable mathematical object whose signatures are written plainly in 703.1 megabases of real genomic data.

Future work will relax truncation, scale to thousands of genomes in parallel, and explore whether the axioms can be derived from first principles in information geometry or algebraic topology.